
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

Anonymous Author(s)

Affiliation

Address

email

Abstract

For all the overall fidelity of the images they produce, generative image models trained on large-scale datasets systematically fail to render seemingly elementary concepts—e.g., human hands or objects appearing in groups of four—despite the fact that these concepts are abundantly and faithfully represented in the very corpora on which the models are trained. Such failures have hitherto been catalogued only anecdotally, leaving unresolved the central question of whether they are isolated idiosyncrasies or the surface manifestations of a structural limitation intrinsic to models trained on internet-scale corpora. We resolve this question by introducing a systematic and fully unsupervised methodology that identifies and characterizes “conceptual blindspots”—concepts that are present in the training distribution yet absent from, or systematically distorted in, a model’s generations. Our method leverages sparse autoencoders (SAEs) to extract interpretable concept embeddings and, through them, to quantify and directly compare the prevalence of each individual concept between real and generated images. As the representational backbone of this comparison, we train an archetypal SAE (RA-SAE) on DINOv2 features spanning 32,000 concepts—the largest such SAE reported to date—which furnishes the concept-level resolution required to localize precisely where a model’s output distribution departs from the natural image distribution. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), our approach pinpoints concrete suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees), and it establishes that suppression is the dominant failure mode—one shared systematically across every model we evaluate. Descending to the level of individual datapoints, the same framework isolates memorization artifacts—instances in which models reproduce highly specific visual templates encountered during training. Together, these results establish a theoretically grounded, model-agnostic framework that diagnoses generative image models by measuring the conceptual fidelity of their outputs against the underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have attained unprecedented capabilities, driving applications both within the vision domain [1–9] and well beyond it [9–15]. This success notwithstanding, a growing body of qualitative [16–18] and quantitative [19–20] evidence establishes that these models routinely fail to render even elementary concepts—among them human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. Tellingly, when prompted for such concepts, these models do not omit them outright but instead synthesize closely related yet incorrect structure—rendering, for



Figure 1: $\delta(k)$ quantifies a model’s tendency to over- or under-generate a concept c_k relative to its frequency in the natural data. Concepts with $\delta(k) < 0.1$ are designated **suppressed conceptual blindspots** and those with $\delta(k) > 0.9$ **exaggerated conceptual blindspots**. The images shown, produced by four widely used generative image models, juxtapose concepts that fall into each blindspot regime against concepts the same models render faithfully, rendering the two failure regimes directly legible at a glance.

instance, hands with six fingers. We term these failure modes “conceptual blindspots”¹ and their existence is genuinely counterintuitive: the training corpora contain abundant, faithful demonstrations of the very concepts that the models fail to reproduce. This tension frames the central question of our work: are such failures the idiosyncratic quirks of a handful of concepts, or the signature of a systematic phenomenon in which, across a broad spectrum of concepts, models are consistently either unable or overly inclined to generate them? We establish that the latter holds.

Existing approaches for evaluating generative image models are structurally ill-suited to answering this question [28, 29]. The prevailing metrics are coarse-grained scores of image realism—chief among them FID [30]—which aggregate over entire images and are, by construction, blind to the concept-level distributional failures at issue. Diversity- and coverage-oriented metrics such as CLIPScore [31–33] move closer, yet they too operate at the granularity of whole images or prompts rather than that of individual features or concepts [34, 35], and are therefore constitutively unable to localize *which* concepts a model fails to reproduce. Qualitative protocols—participant surveys [5, 17, 36–38] and open-ended exploration [39]—can surface individual conceptual failures, but they fall short of a scalable, repeatable methodology that transfers across model families and supports systematic, quantitative comparison.

This work. We close this gap by developing an automated, unsupervised methodology that identifies concepts in the data distribution whose odds of being generated by the trained model diverge from their odds under the true data-generating process. Crucially, rather than merely quantifying aggregate distribution coverage, the methodology pinpoints *which* concepts constitute the mismatch and adjudicates, for each, whether it is suppressed or exaggerated. Our contributions are three-fold:

- **A formalization of conceptual blindspots in generative image models.** We cast conceptual blindspots as measurable discrepancies between the concept content of natural images and that of a model’s generations, and we derive from first principles a quantitative measure of these discrepancies grounded in the data-generating process (Section 3). This formalization supplants anecdotal, human-defined concept evaluations with a single, calibration-free criterion that we prove both concentrates rapidly from modest sample sizes and preserves the ordering of the underlying energy gap—two properties that together establish the measure as a reliable and theoretically sound diagnostic.
- **A scalable, unsupervised method for detecting conceptual blindspots with sparse autoencoders.** We operationalize this formalization as a concrete, model-agnostic pipeline that uses sparse autoencoders (SAEs) to extract and directly compare concept distributions between natural and synthesized images (Section 4). To this end, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features spanning 32,000 concepts—the largest such RA-SAE reported to date—which delivers the concept-level resolution the analysis demands while yielding seed-independent, reproducible concept bases.
- **An exploratory tool and a multi-level empirical analysis.** We release an interactive web tool that supports distribution-, datapoint-, and composition-level analysis of blindspots across models (Sections 5.1–5.4), and we apply our method to four widely used models—Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5). The analysis surfaces concrete suppressed

¹We borrow the term “blindspots” from the psychology literature [27], in which it denotes scenarios where an agent makes biased decisions despite exposure to observations that contradict the rationale behind those decisions.

conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), illustrated in Fig. 1 and attributes them to shared training-data properties and model-specific inductive biases alike.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early work in explainable AI, computer vision included, centered on attributing predictions to influential input regions [40–42]. Such attribution methods, however, convey little semantic information about the learned representations and can yield explanations that are demonstrably unfaithful [43–45]. Concept-based interpretability [46] emerged precisely to overcome these shortcomings, identifying semantically meaningful directions in a network’s representation space and thereby revealing not merely where a model attends but which concepts and structures it encodes [47–49].

Recent work establishes that the leading concept-based interpretability methods—ACE [44], CRAFT [48], and sparse autoencoders (SAEs) [50–51]—all instantiate a single underlying dictionary-learning problem under differing constraints [52]. Among them, SAEs constitute the most scalable route to concept-based interpretability. While the original formulation exhibits well-documented limitations—overly specific features [53], compositionality challenges [54], and limited intervention effects [55]—subsequent refinements address these directly, most notably the archetypal SAE (RA-SAE) [56] that we adopt and scale here.

Complementary interpretability techniques—prompt-based probing [57], attention-map and activation visualizations [58–59], and dataset-level statistics such as diversity or distribution-coverage metrics [32–33]—each afford only partial purchase on our question. Their limitation is a common and decisive one of granularity: they operate over whole images or prompts rather than over the fine-grained features and concepts that blindspot analysis requires [34–35]. They rely, moreover, on subjective interpretation and do not disambiguate distinct failure modes [60]. Existing methods and metrics are thus inadequate for systematically localizing the feature- and concept-level weaknesses of generative image models [28]—precisely the gap our framework is designed to close.

2.2 Generative Image Models

Diffusion-based methods now dominate generative vision modeling across modalities, spanning image [4, 61–64], video [65–72], and 3D [7, 73–75] synthesis. In image generation, the lineage traces to denoising diffusion probabilistic models (DDPMs) [76], subsequently generalized to non-Markovian diffusion processes by denoising diffusion implicit models (DDIMs) [77].

The Stable Diffusion (SD) [9] family brought DDPMs to the research and open-source communities at scale and was extended through successive releases—SD 2 [78], SD 3 [79], and SD XL [80]. A large ecosystem of architectural modifications followed, imposing additional constraints on the diffusion process—over style [81–82], pose [83], and identity [84–85], among others—and admitting alternative input modalities such as image-to-image generation. A parallel line of latent diffusion models built atop SD—Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89]—has proliferated in turn. It is precisely this architectural heterogeneity that our model-agnostic methodology is built to compare on a common footing, without recourse to model-specific instrumentation.

2.3 Datasets for Generative Image Models

The recent success of generative vision models rests largely on the confluence of abundant computation and large-scale internet datasets [90–92]. LAION-5B [93] has been pivotal to the training of open-source text-to-image models, SD and its derivatives among them. Scraped from Common Crawl [94], it comprises over 5B image-caption pairs, 2.3B of them in English; other prominent corpora include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M pairs, respectively.

The dataset’s prominence has been accompanied by mounting concerns over its biases [97–101]. Filtering efforts notwithstanding, harmful content persisted [97–99]—including NSFW material [100]

and hundreds of CSAM instances [101]—prompting the corpus’s temporary withdrawal from official channels. The data is further marked by low-quality images [102] and internet-style captions (e.g., product descriptions) that diverge sharply from how users actually prompt the trained models [103]. These distributional properties of the training corpus are, we argue, a direct cause of the conceptual blindspots our analysis is designed to expose, and it is exactly their downstream effects that our framework renders measurable.

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks sparse representations in which each sample is reconstructed as a linear combination of a small number of dictionary atoms [104–106]. Grounded in compressed sensing theory [107–108], the field has progressed from early vector quantization [109] to a rich family of formulations, encompassing non-negative matrix factorization [110–111], sparse PCA [112], and K-SVD [113]. More recent developments span online algorithms [114], structured sparsity [115], and theoretical recovery guarantees [116–117], together with increasingly tight connections to deep learning [118–119].

Sparse coding has, in turn, been brought to bear on the emergence of high-level concepts inside diffusion models [120]. Prior to diffusion models, concept-grounded interpretability was applied to earlier generative architectures through concept-bottleneck models, which demand human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. Both paradigms, however, presuppose a fixed, human-defined concept vocabulary and are therefore constitutively incapable of registering trends the user has not anticipated in advance—the very trends that an unsupervised method is uniquely positioned to surface.

For the evaluation of generative image models specifically, aggregate metrics—precision, recall, density, and coverage [35–123]—and latent density scores that predict sample quality from the model’s latent space [124] have been proposed. These reliably expose distributional gaps in aggregate, yet they remain silent on the decisive question of which specific concepts are under- or over-represented.

These limitations converge on a single unmet need: a scalable, unsupervised framework that systematically identifies and quantifies concept-level failure modes in generative image models [125]—the framework we develop next.

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and that of a model’s generations. This formulation yields a principled, quantitative measure that identifies precisely which concepts a generative model under- or over-represents relative to its data distribution; the overall process is illustrated in Fig. 2. While the derivation invokes standard assumptions from the identifiability and disentanglement literature [126–130], we demonstrate empirically that the framework elicits meaningful phenomenology even when those assumptions hold only approximately—establishing that its diagnostic value is robust to their exact satisfaction rather than contingent upon it.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(\mathbf{c}) = \exp(-E(\mathbf{c}))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(\mathbf{c}) = \sum_k E(c_k)$ and hence $p(\mathbf{c}) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $\mathbf{c} \in \mathcal{C}$ to observations $\mathbf{x} \in \mathcal{X}$, i.e., $\mathbf{x} = G(\mathbf{c})$.

For notational economy, we let $p(\cdot)$ denote both the latent density $p(\mathbf{c})$ and its pushforward to image space $p(\mathbf{x})$, where $\mathbf{x} = G(\mathbf{c})$. It is the invertibility of G that legitimizes this overloading: through the change-of-variables formula, it induces a well-defined distribution over $\mathcal{X}$. The individual dimensions of the latent space thus constitute the **Concepts** underlying the data distribution $P_{\mathcal{X}}$, defined over the observation space of images $\mathcal{X}$. Concretely, distinct latents may encode concepts such as color, shape, size, and location [131–132]. We additionally let the data-generating process associate a text description $t \in \mathcal{T}$ with each sampled im-

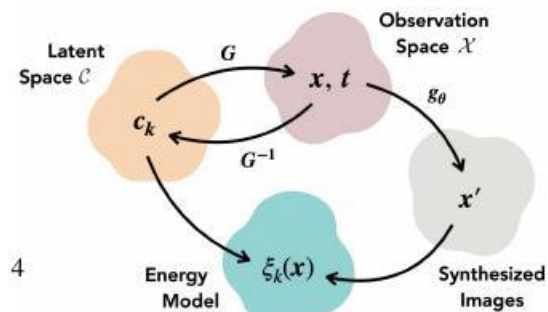


Figure 2: **Schematic of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations $(\mathbf{x}, t)$

age, though we leave this association implicit rather than modeling it explicitly. These descriptions serve as the conditioning signal for a text-conditioned **Generative image model** g_θ with parameters θ : trained on a corpus of image-text pairs, g_θ maps a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ together with a scene description $\mathbf{t}$ to an image $\mathbf{x}$ that realizes that description.

Defining conceptual blindspots in g_θ therefore reduces to a single, well-posed comparison: the probability mass a concept receives under the data-generating process against the mass it receives under the model. To operationalize this comparison, we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and construct a matched set of synthesized images $\mathcal{D}'_\mathcal{X}$ from the identical text descriptions. We estimate a concept’s probability of occurrence through an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to a d -dimensional vector whose k^{th} coordinate is the scalar energy of the k^{th} concept, i.e., $\xi_k(\mathbf{x}) = E(c_k)$. It is, once again, precisely the invertibility of the data-generating process that renders these estimates recoverable. The probability the process assigns to an image $\mathbf{x}$ then follows as $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (with Z_ξ the partition function), which aggregates to the population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{\mathbf{x} \in \mathcal{D}_\mathcal{X}} p_k(\mathbf{x})$. Composing this with the sigmoid map $\sigma(\cdot)$, we now define the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$.

Definition 2 (Energy Difference). Let $\mathbf{x} \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-generating process G , and let $\mathbf{x}' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

The energy difference in the k^{th} concept thus expresses the relative probability that the concept occurs in a target set of observations (here, $\mathcal{D}'_\mathcal{X}$) against a baseline set (here, $\mathcal{D}_\mathcal{X}$), normalized to $(0, 1)$ by the sigmoid so that 0.5 denotes exact parity. This single scalar furnishes a direct and interpretable criterion for conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

In sum, a conceptual blindspot is a concept whose likelihood of occurrence in generated images departs markedly—through either suppression or exaggeration—from its prevalence under the data-generating process. Suppressed concepts activate disproportionately rarely ($\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are systematically overrepresented ($\delta(k) > \lambda_{\max}$); throughout, we set $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to demarcate these two regimes.

This definition at once generalizes and sharpens the classical notion of “mode collapse” studied in past work (e.g., Bau et al. [39]), from which it differs chiefly in granularity. Whereas mode collapse concerns the suppressed or exaggerated odds of generating *entire images*, our measure isolates the shifted odds of *individual concepts*. A model’s failure to produce objects on a `white` background, for instance, registers in our formalism as a suppressed conceptual blindspot localized to that precise concept, rather than being obscured within a global, image-level collapse.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We now instantiate the formalization of Sec. 3 as a concrete, reproducible pipeline for identifying conceptual blindspots in a generative model g_θ . The construction requires exactly two ingredients:

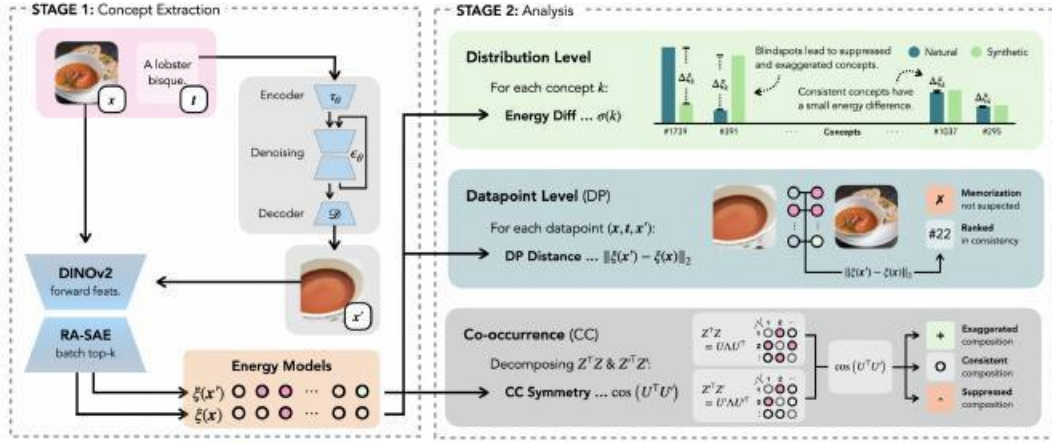


Figure 3: **Concept extraction pipeline.** For a triplet (x, t, x') , concepts are extracted in two sequential stages: we first compute the DINOv2 features of each image and then map them through the RA-SAE into sparse concept embeddings, which constitute the energy models $\xi(x)$ and $\xi(x')$.

By construction, the k^{th} coordinate $\xi_k(x) = E(c_k)$ holds the energy of the k^{th} concept, so that real and generated images become directly comparable within a shared concept basis.

(i) a set of images sampled from g_θ that admit a one-to-one correspondence with samples from the ground-truth generative process, and (ii) an energy model that renders the two sets comparable within a common concept basis. As to notation, $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_0$ the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ denotes element-wise non-negativity. For $n > 0$, we write $[n] := \{1, \dots, n\}$ and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

From Prompts to Latent Representations. To identify conceptual blindspots in g_θ , we compare a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs (x, t) drawn from the data-generating process $\mathcal{G}$ against their synthetic counterparts, generated by g_θ from the identical text descriptions, so that each real image stands in exact correspondence with a prompt-aligned generation.

Concretely, given a caption t , we synthesize the counterpart image x' with a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, instantiated as a denoising diffusion probabilistic model (DDPM) [78][86][87]. Sampling proceeds in latent space along the reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The terminal latent γ_0 is decoded by a pretrained VAE to yield the synthetic image $x' = \text{VAE}(\gamma_0)$. For the remainder of the paper we treat g_θ as a black box mapping prompts to images, $t \mapsto x'$; it is precisely this deliberate abstraction that renders the method architecture-agnostic and applicable, without modification, across every diffusion model we study.

Defining the Energy Model. Prior work establishes that self-supervised learning methods can invert the data-generating process and recover its underlying energy function up to a linear transformation [126][128][133][134]; we exploit this property directly, adopting DINOv2 [135], a state-of-the-art self-supervised encoder, as the backbone of our energy model. Because the number of concepts underlying the DGP is expected to exceed the dimensionality of the encoder’s feature space [51][136], these concepts are of necessity represented in superposition; we therefore train sparse autoencoders (SAEs) on the DINOv2 features to disentangle the subspaces corresponding to individual concepts [50][56][137–139]. The rationale is exact: if DINOv2 encodes the generative concepts along approximately orthogonal directions—consistent with the independence structure posited in Def. 1—then an SAE trained on its features isolates each concept along a distinct latent dimension [136]. The activation the SAE assigns to that dimension then serves as our estimator of the ground-truth energy of the concept it encodes.

Formally, let $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the feature extractor (the DINOv2 model); we compute features $\mathbf{a} = \mathbf{f}(x) \in \mathbb{R}^d$ for both the natural and the synthetic images in $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$. Writing n for the number of image-text pairs, we stack the real and generated features into matrices $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$. Each feature vector is subsequently decomposed by an SAE into a sparse combination of learned concept atoms. Let $\mathbf{D} \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors and $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ the SAE encoder that maps features to sparse codes. Applying Ψ row-wise yields the activation matrix

264 $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$, whose i -th row $\mathbf{z}_i = \Psi(\mathbf{a}_i)$ constitutes the concept decomposition of the
 265 i -th image. The decoder reconstructs features through $\mathbf{Z}\mathbf{D}^\top$, and the SAE is trained to minimize
 266 reconstruction error subject to sparsity and non-negativity constraints:

$$\min_{\Psi, \mathbf{D}} \|\mathbf{A} - \Psi(\mathbf{A})\mathbf{D}^\top\|_F^2 \quad \text{s.t.} \quad \Psi(\mathbf{A}) \geq 0, \|\Psi(\mathbf{A})_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

267 Vanilla SAEs are prone to converging on arbitrarily oriented dictionaries, a pathology that renders
 268 downstream analyses acutely sensitive to the random seed. To eliminate this instability and thereby
 269 guarantee seed-independent, reproducible results, we employ the *Archetypal* SAE (RA-SAE) [56]
 270 under a TOP-K sparsity constraint [137]. RA-SAE constrains each dictionary atom to be a convex
 271 combination of the training data. Formally, we write $\mathbf{D} = \mathbf{W}\mathbf{A}$ with $\mathbf{W} \in \Omega_{K', n}$, the set of
 272 row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

273 Consequently, every atom $\mathbf{D}_i$ lies within the convex hull of the data $\text{conv}(\mathbf{A})$, and any reconstruction
 274 $\mathbf{Z}\mathbf{D}^\top$ resides within the conic hull $\text{cone}(\mathbf{A})$. This constraint tethers the learned concepts to the
 275 support of the data distribution and thereby guarantees that they remain faithful to it [56]. Once
 276 trained, the SAE yields a single, consistent concept basis in which both real and generated images are
 277 jointly expressed: $\mathbf{Z}$ for the real images and $\mathbf{Z}'$ for their generated counterparts. *Because both sets of*
 278 *codes are expressed over the same interpretable concepts, they render prompt-conditioned visual*
 279 *semantics directly comparable, and each concept’s activation supplies precisely the energy value on*
 280 *which our blindspot analysis operates.* Our method therefore constitutes a structured pipeline that,
 281 from a prompt and its associated real image $(t, \mathbf{x})$, returns two sparse concept vectors $(\mathbf{z}, \mathbf{z}')$ and, in
 282 doing so, situates the real and generated visual content within a common conceptual basis for direct,
 283 concept-level comparison.

284 To summarize the full pipeline: beginning from a captioned image $(t, \mathbf{x})$, we synthesize a counterpart
 285 $\mathbf{x}'$ and embed both images into a shared sparse concept space through the vision encoder and
 286 the trained SAE, obtaining interpretable representations $(\mathbf{z}, \mathbf{z}')$ that constitute the foundation for
 287 quantifying the conceptual shifts the generative process induces.

288 5 Results

289 We analyze four widely used generative image models trained on LAION-5B—SD 1.5,
 290 SD 2.1, PixArt, and Kandinsky—over $|\mathcal{X}| = 10,000$ image-text pairs and their
 291 prompt-matched generations (Fig. 4). The analysis unfolds across three complemen-
 292 tary and mutually reinforcing levels (Fig. 3): a ● **distribution-level** evaluation reveals
 293 suppressed and exaggerated concepts; a ● **datapoint-level** analysis surfaces failures at-
 294 tributable to ambiguity, omission, and memo-
 295 rization; and a ● **compositional** analysis ex-
 296 poses structured distortions in the geometry of
 297 concept co-occurrence.

299 At the center of our contribution stands an in-
 300 teractive exploratory tool, shown in Fig. 5: a
 301 web-based interface organized around a UMAP
 302 projection of concept representations that sup-
 303 ports the visualization and direct comparison of
 304 concept-level energy differences. It is publicly
 305 available at a software repository, together with pre-computed energy-difference data for all four
 306 evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky), and every analysis reported below is
 307 derived through it. Its principal capabilities are three:

- 308 • **Contrasting models and architectures.** For each evaluated model, the tool renders a
 309 UMAP visualization spanning all 32,000 RA-SAE concepts, every scatter point representing
 310 an individual concept color-coded by its energy difference, so that entire blindspot profiles
 311 are compared across architectures at a glance.



Figure 4: **Representative image-caption pairs** from LAION-5B, each displayed alongside the images synthesized from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

- **Inspecting concepts.** Each concept is documented by a card that reports its key statistics, its most representative real and generated images (x , x'), and its co-occurrence patterns.
- **Exploring blindspots.** Beyond the UMAP and per-concept views, the tool furnishes global rankings of suppressed and exaggerated blindspots, foregrounding the most pronounced conceptual blindspots for immediate inspection.

Rather than exhaust any single phenomenon, we report a set of high-level findings that together demonstrate the breadth of the tool and open the way to broader, user-directed exploration.

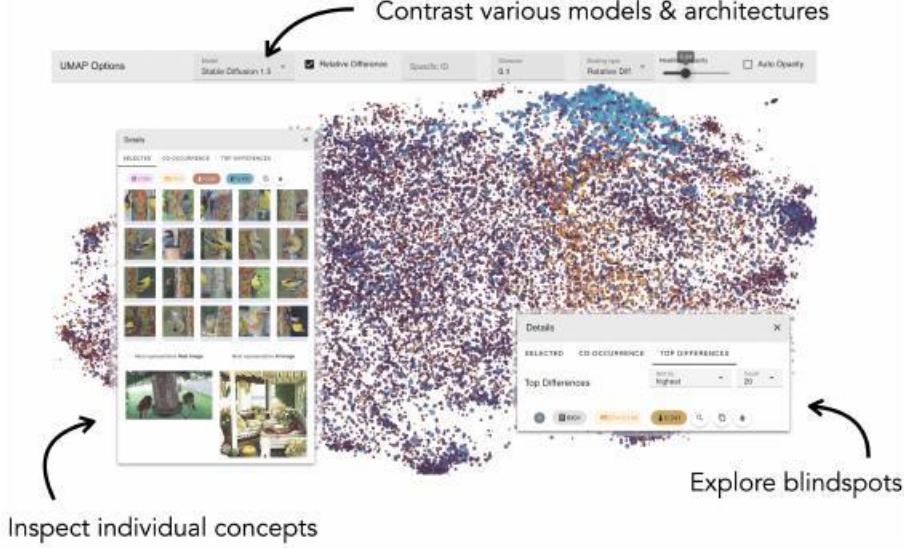


Figure 5: **Overview of the exploratory tool.** The web interface displays a per-model UMAP projection in which each dot is a concept color-coded by its energy difference. Selecting a concept opens a detail panel that reports its statistics and its most representative natural and generated images, x and x' , together with an ordered list of the concept’s co-occurrences and the global blindspot rankings, so that distribution- and datapoint-level evidence may be interrogated within a single unified interface.

5.1 Generative Image Models Cannot Produce Unseen Concepts

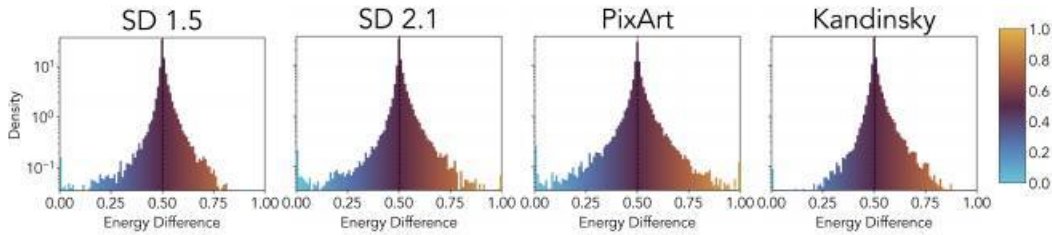


Figure 6: **Concept energy distribution.** Log-scale histograms of the energy difference $\delta(k)$ over all 32,000 concepts, contrasting the natural and synthesized distributions for each evaluated model. Mass to the left corresponds to suppressed (under-represented) concepts and mass to the right to exaggerated (over-represented) concepts; the pronounced left skew common to all four histograms is direct evidence of a systematic, cross-model bias toward suppression.

To characterize the disparities between the generative models and the natural image distribution, we first analyze the marginal energy difference $\delta(k)$ across all 32,000 concepts learned by the RA-SAE. As defined in Sec. 4, this quantity measures the relative prevalence of each concept in the synthesized against the natural image set: $\delta(k) < 0.1$ marks concept k as under-represented (suppressed) in the generated images, and $\delta(k) > 0.9$ marks it as over-represented (exaggerated). Fig. 6 presents the distribution of $\delta(k)$ for each of the four models. Every model exhibits a heavy-tailed histogram carrying substantial mass at both extremes, which establishes that concept-coverage

discrepancies are systematic rather than incidental. Decisively, the left tail—the suppressed concepts—is consistently denser and longer than the right across all four models, evidence of a robust and shared tendency toward concept suppression. This asymmetry is quantified directly by the uniformly negative skewness of the four distributions: -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

While all four models thus display both suppressed and exaggerated concepts, their detailed profiles diverge in informative ways. PixArt, for instance, exhibits the widest spread, consistent with a more heavily suppressed concept distribution. The left skew common to every distribution nonetheless attests to a shared propensity for concept omission that transcends any single architecture; the finer structure of this behavior is the concern of the sections that follow.

5.2 Structure and Specificity of Conceptual Blindspots

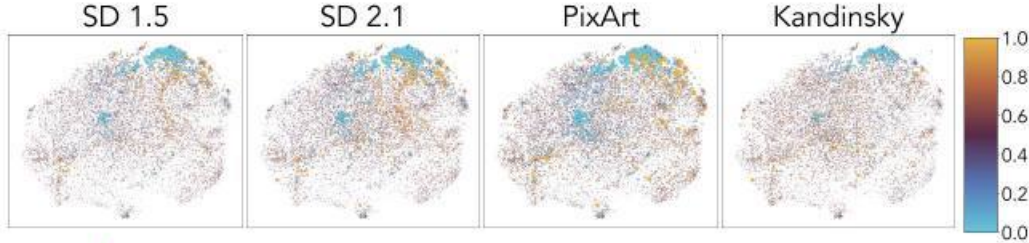


Figure 7: **Structure of concept energy differences.** UMAP visualizations of the 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions. The emergence of coherent clusters—rather than scattered points—demonstrates that conceptual blindspots are spatially organized in concept space, with suppressed concepts concentrated at the blue end and exaggerated ones at the yellow end.

Whereas the previous section quantified the marginal discrepancies in concept frequency, we now characterize their global structure, embedding all 32,000 concepts into two dimensions via UMAP on the sparse codes and again coloring each concept by its $\delta(\cdot)$ value. As Fig. 7 shows, distinct concept clusters emerge for every model. These clusters frequently coincide with contiguous regions of conceptual blindspots—most conspicuously for suppressed (blue) concepts—establishing that blindspots are not randomly distributed but spatially structured, and hence traceable to shared biases in the training distribution or to architectural priors. To determine how consistent these blindspot patterns are across models, we analyze both the magnitude and the structure of the concept-level $\delta(\cdot)$ values. Fig. 8 reports scatter plots and pairwise Pearson correlations between the $\delta(k)$ vector of SD 1.5 and those of the remaining models. SD 1.5 and SD 2.1 correlate strongly ($r = 0.82$), exactly as their shared architecture and training pipeline would predict. Their correlations with PixArt and Kandinsky, by contrast, are markedly lower ($r = 0.41$ and $r = 0.46$, respectively), establishing that these architectures allocate their blindspots to substantially different regions of the conceptual space.

Taken together, this analysis establishes that blindspots occupy two distinct regimes: some are universally shared—most plausibly a consequence of dataset properties—whereas others are strongly model-specific, arising from idiosyncrasies of training dynamics or model capacity. It is this dichotomy that directly motivates the separate treatment, in the sections that follow, of blindspots shared across models and those unique to individual models.

5.3 Qualitative Examples of Blindspots

We now examine concrete instances of both suppressed and exaggerated blindspots, so as to characterize the concepts that populate each regime. Fig. 9a presents a blindspot suppressed by every model: all four evaluated models fail to reproduce the concept `solid white on documents`. Although the caption references this concept explicitly, none of the generations recover the intended visual semantics—direct evidence that this region of concept space is systematically under-sampled across the entire set of models. Fig. 9b, by contrast, isolates a model-specific blindspot: the concept `pan` is rendered accurately by three models yet is conspicuously absent from Kandinsky’s generations. Together, these two cases concretely instantiate the dichotomy established in Sec. 5.2 wherein cross-model agreement proved high for some concepts and limited for others.

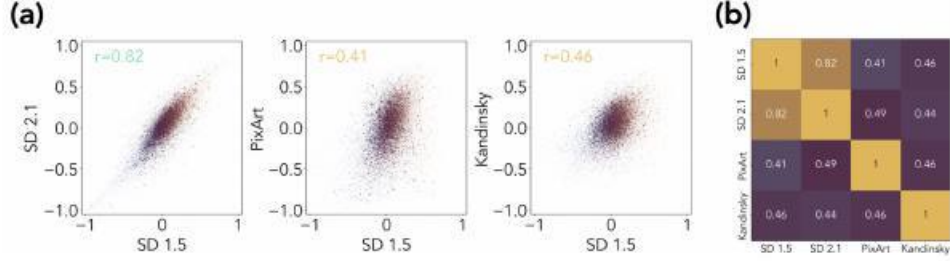


Figure 8: **• Cross-model concept energy correlation.** Pairwise scatter plots of $\delta(k)$ across the four evaluated models, with Pearson correlation coefficients reported at top left. The strong alignment between SD 1.5 and SD 2.1 ($r = 0.82$) stands in sharp contrast to the substantially weaker correlations with the other architectures ($r = 0.41$ and $r = 0.46$), quantifying precisely the extent to which blindspots are model-specific. **• Correlation matrix of conceptual blindspots.** Heatmap of the pairwise Pearson correlations of δ across all models, summarizing how much of each model’s blindspot profile is shared with the others.

Conversely, Fig. 10 presents a case of exaggeration, in which the concept shadow under `animal` is over-emphasized across generations. Although each shadow is individually plausible, their uniformly pronounced rendering across models—set against their subtler and more variable appearance in natural images—betrays an overactive prior. Notably, a deliberate search for concepts exaggerated by a single model alone yielded no clear examples, indicating that, unlike suppression, exaggeration is approximately universal across the evaluated models.

Collectively, these examples make concrete how conceptual blindspots surface in generated outputs and confirm that our energy-based diagnostic reliably resolves both shared and model-specific failure modes. In particular, it exposes otherwise surprising limitations—such as the consistent, cross-model failure to reproduce clear or solid background elements like whitespaces on documents—and thereby establishes that certain failure patterns originate in architectural constraints or training-data biases that transcend the idiosyncrasies of any single model.

Informative as these aggregate analyses are, they pose a sharper question: do blindspots arise only in aggregate over many samples, or do they persist at the level of in-



Figure 10: **• Example of an exaggerated conceptual blindspot.** Four synthesized images D'_x , archetypal for the concept shadow under `animal`, displayed alongside the corresponding natural image D_x and the caption t used to gen-

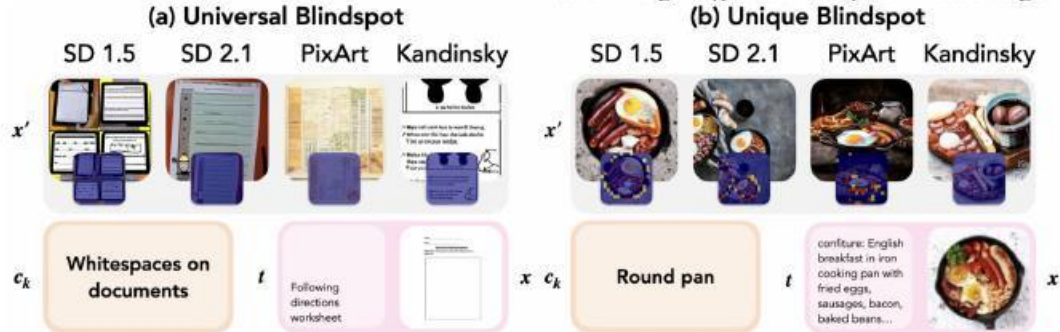


Figure 9: **• Examples of suppressed conceptual blindspots.** Each natural image x , its representative concept c_k , and its caption t are displayed alongside the four synthesized images x' produced by S_θ . The universal blindspot (solid white on documents) manifests in all four evaluated models, whereas the model-specific blindspot (pan) manifests in Kandinsky alone.

dividual datapoints? Addressing this question probes the mechanisms behind blindspots directly, distinguishing prompt misinterpretation from latent memorization and pinpointing where the two coincide.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine the individual natural-generated image pairs whose $\delta(\cdot)$ values, averaged over all concepts, are largest and smallest—the two extremes at which the diagnostic is at its most informative. This datapoint-level view cleanly separates model successes from failures, the latter of which we find to arise predominantly from either prompt ambiguity or memorization artifacts. Fig. 11a shows pairs with near-zero average $\delta(\cdot)$, in which the generated image is conceptually indistinguishable from the original. Qualitative inspection reveals that this agreement stems not from faithful capture of the prompt semantics but from verbatim replication of memorized templates: the recurrence of near-identical visual structures (e.g., clothing outlines or object arrangements) indicates that the model is reproducing over-frequent patterns lifted directly from the training data. Fig. 11b, conversely, shows pairs among the largest $\delta(\cdot)$, which signal substantial conceptual divergence between the synthesized and natural images. Some of these discrepancies trace to underspecified or noisy captions, but others expose genuine blindspots: the prompt describes a clear concept, faithfully present in $\mathcal{D}_{\mathcal{X}}$, that the model nonetheless fails to realize in $\mathcal{D}_{\mathcal{X}'}$. Such cases demonstrate that even when language grounding is fully adequate, certain concepts lie beyond the model’s generative reach.



Figure 11: • **Datapoint-level conceptual alignment.** (a) Pairs with minimal energy difference, in which the model reproduces memorized training patterns. (b) Pairs with large energy difference, in which pronounced conceptual divergence arises either from prompt ambiguity or from genuine limitations of the model.

5.5 Analyzing Post-Training Effects

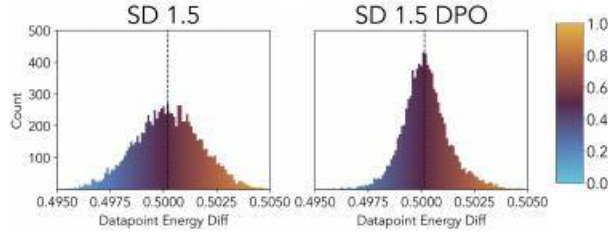


Figure 12: **Effect of DPO on concept fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distributions for SD 1.5 with and without DPO. The DPO variant shifts mass toward smaller differences, direct evidence of improved conceptual alignment with the natural distribution.

Post-training protocols such as safety fine-tuning have been argued to reduce the diversity of model generations [140]. Because our pipeline isolates precisely the differences between a model’s generations and the ground-truth DGP, it constitutes a natural and direct instrument for quantifying the effect of DPO—a widely used safety fine-tuning protocol [141]. We compare two variants of SD 1.5, one trained with DPO and one without. For each image pair $(\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X})$ we compute the ℓ_2 distance between their concept energy vectors, $\|\xi(\mathcal{D}'_\mathcal{X}) - \xi(\mathcal{D}_\mathcal{X})\|_2$. Fig. 12 presents the resulting histogram of datapoint-wise energy differences. The DPO variant attains both a lower median and a narrower spread, demonstrating that its generated concepts align more consistently with the ground-truth DGP. DPO thus acts to regularize the model’s concept distribution, steering its outputs toward the semantic content of the observed inputs. While our analysis does not disentangle the specific inductive biases DPO introduces, it furnishes direct empirical evidence that an objective favoring human-preferred generations, as a by-product, also improves agreement with the training distribution. Crucially, this improvement is bidirectional: DPO attenuates *both* the over- and the under-activation of individual concepts relative to the natural baseline. These findings underscore the value of our pipeline for characterizing post-training interventions—not merely through output quality, but through the reshaping of the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

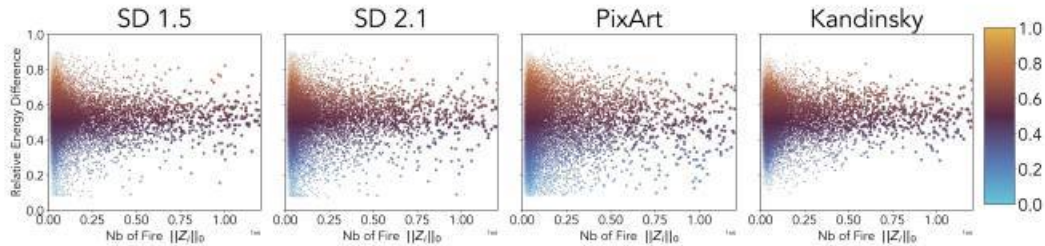


Figure 13: **Concept fidelity across the frequency spectrum.** Scatter plots relating concept frequency (x-axis) to energy difference (y-axis) for the four evaluated models. Each point is a concept, its size proportional to its activation frequency; for the trend, consistent across all four models, confirms that low-frequency, long-tail concepts concentrate the largest fidelity errors.

We hypothesized above that conceptual blindspots are not solely architectural artifacts but may arise directly from the distributional properties of individual features. Here we put this hypothesis to an empirical test, asking whether concepts that are rarely activated in natural images—those in the long tail of the data distribution—systematically induce blindspots in generative models. We process the natural dataset $\mathcal{D}_\mathcal{X}$ through the trained SAE and, for each concept k , compute its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$, the count of non-zero activations of concept i across all images. We then relate this frequency to the absolute energy difference observed in the generated outputs. Fig. 13 visualizes this relationship for all four models. A clear and consistent trend emerges: high-frequency concepts exhibit small energy discrepancies, whereas rare concepts—particularly suppressed ones ($\delta(k) < 0.5$)—incur substantial alignment errors. This establishes that many blindspots are not random model quirks

454 but the systematic downstream consequence of long-tailed concept distributions, and it implies that
455 mitigating them will demand not architectural changes alone but also data-level interventions such as
456 reweighting or augmentation.

457 6 Discussion

458 Our analysis uncovers numerous conceptual blindspots across four widely used generative image
459 models, and it does so through a single, unified framework. These results, moreover, only begin
460 to exploit the framework’s reach: each individual finding could sustain a dedicated investigation
461 of its own. Rather than pursue any single thread in depth, we have deliberately demonstrated the
462 breadth of our method and exploratory tool, which together support—out of the box—the systematic
463 identification of concepts a model struggles to generate, the detection of memorization artifacts, the
464 discovery of datapoints with insufficient captions, the quantification of post-training effects, and
465 the characterization of conceptual shifts across model architectures. This breadth establishes the
466 framework as a general foundation on which follow-up work can build, whether by deepening the
467 analysis, widening the set of evaluated architectures, or investigating the root causes of conceptual
468 blindspots. A particularly promising direction is to endow the RA-SAE with hierarchical concept
469 representations, which would enable a correspondingly more nuanced, compositional analysis.

470 **Limitations.** Several limitations delimit the scope of our conclusions and, in doing so, clarify pre-
471 cisely what the framework does and does not establish. Because concept extraction and representation
472 rest on DINOv2 and the RA-SAE, our analysis is confined to the concepts these models encode:
473 concepts they represent poorly will of necessity escape detection. Furthermore, while a sample of
474 10,000 images is substantial—and, as our concentration analysis rigorously establishes, sufficient
475 for tight estimation of δ —it may not exhaust the long tail of rare concepts, nor fully resolve concept
476 co-occurrence and other higher-order compositional statistics.

References

- [1] OpenAI. Sora: Creating video from text, 2024. URL <https://openai.com/sora/>
- [2] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4195–4205, 2023.
- [3] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International conference on machine learning*, pages 8821–8831. PMLR, 2021.
- [4] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems*, 35:36479–36494, 2022.
- [5] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. GLIDE: Towards photorealistic image generation and editing with text-guided diffusion models. *arXiv:2112.10741*, 2021.
- [6] Chen Wang, Jiatao Gu, Xiaoxiao Long, Yuan Liu, and Lingjie Liu. Geco: Generative image-to-3d within a second. *arXiv preprint arXiv:2405.20327*, 2024.
- [7] Ben Poole, Ajay Jain, Jonathan T Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv preprint arXiv:2209.14988*, 2022.
- [8] Elad Richardson, Gal Metzer, Yuval Alaluf, Raja Giryes, and Daniel Cohen-Or. Texture: Text-guided texturing of 3d shapes. In *ACM SIGGRAPH 2023 conference proceedings*, pages 1–11, 2023.
- [9] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.
- [10] Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, et al. Do as i can, not as i say: Grounding language in robotic affordances. *arXiv preprint arXiv:2204.01691*, 2022.
- [11] Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, et al. Inner monologue: Embodied reasoning through planning with language models. *arXiv preprint arXiv:2207.05608*, 2022.
- [12] Wenlong Huang, Pieter Abbeel, Deepak Pathak, and Igor Mordatch. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In *International Conference on Machine Learning*, pages 9118–9147. PMLR, 2022.
- [13] Yun-Chun Chen, Selena Ling, Zhiqin Chen, Vladimir G Kim, Matheus Gadelha, and Alec Jacobson. Text-guided controllable mesh refinement for interactive 3d modeling. In *SIGGRAPH Asia 2024 Conference Papers*, pages 1–11, 2024.
- [14] Lei Zhong, Yiming Xie, Varun Jampani, Deqing Sun, and Huaizu Jiang. Smoodi: Stylized motion diffusion model. In *European Conference on Computer Vision*, pages 405–421. Springer, 2024.
- [15] Yawar Siddiqui, Antonio Alliegro, Alexey Artemov, Tatiana Tommasi, Daniele Sirigatti, Vladislav Rosov, Angela Dai, and Matthias Nießner. Meshgpt: Generating triangle meshes with decoder-only transformers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19615–19625, 2024.
- [16] Gary Marcus, Ernest Davis, and Scott Aaronson. A very preliminary analysis of dall-e 2. *arXiv preprint arXiv:2204.13807*, 2022.
- [17] Ángel Alexander Cabrera, Abraham J Druck, Jason I Hong, and Adam Perer. Discovering and validating ai errors with crowdsourced failure reports. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW2):1–22, 2021.

- [18] Rebecca Heigl. Generative artificial intelligence in creative contexts: a systematic review and future research agenda. *Management Review Quarterly*, pages 1–38, 2025.
- [19] Qihao Liu, Adam Kortylewski, Yutong Bai, Song Bai, and Alan Yuille. Discovering failure modes of text-guided diffusion models via adversarial search. *arXiv preprint arXiv:2306.00974*, 2023.
- [20] Colin Conwell, Rupert Tawiah-Quashie, and Tomer Ullman. Relations, negations, and numbers: Looking for logic in generative text-to-image models. *arXiv preprint arXiv:2411.17066*, 2024.
- [21] Wenquan Lu, Yufei Xu, Jing Zhang, Chaoyue Wang, and Dacheng Tao. HandRefiner: Refining malformed hands in generated images by diffusion-based conditional inpainting. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 7085–7093, 2024.
- [22] Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. HanDiffuser: Text-to-image generation with realistic hand appearances. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2468–2479, 2024.
- [23] Qilong Zhangli, Jindong Jiang, Di Liu, Licheng Yu, Xiaoliang Dai, Ankit Ramchandani, Guan Pang, Dimitris N Metaxas, and Praveen Krishnan. Layout-agnostic scene text image synthesis with diffusion models. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7496–7506. IEEE Computer Society, 2024.
- [24] Forouzan Fallah, Maitreya Patel, Agneet Chatterjee, Vlad I Morariu, Chitta Baral, and Yezhou Yang. TextInVision: Text and prompt complexity driven visual text generation benchmark. *arXiv:2503.13730*, 2025.
- [25] Yuefan Cao, Xuyang Guo, Jiayan Huo, Yingyu Liang, Zhenmei Shi, Zhao Song, Jiahao Zhang, and Zhen Zhuang. Text-to-image diffusion models cannot count, and prompt refinement cannot help. *arXiv preprint arXiv:2503.06884*, 2025.
- [26] Colin Conwell and Tomer Ullman. Testing relational understanding in text-guided image generation. *arXiv preprint arXiv:2208.00005*, 2022.
- [27] Mahzarin R Banaji and Anthony G Greenwald. *Blindspot: Hidden biases of good people*. Bantam, 2016.
- [28] George Stein, Jesse Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Ross, Valentin Vilecroze, Zhaoyan Liu, Anthony L Caterini, Eric Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. *Advances in Neural Information Processing Systems*, 36:3732–3784, 2023.
- [29] Boheng Wang, Yunhuai Zhu, Liuqing Chen, Jingcheng Liu, Lingyun Sun, and Peter Childs. A study of the evaluation metrics for generative images containing combinational creativity. *AI EDAM*, 37:e11, 2023.
- [30] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- [31] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7514–7528, 2021.
- [32] Mischa Dombrowski, Weitong Zhang, Sarah Cechnicka, Hadrien Reynaud, and Bernhard Kainz. Image generation diversity issues and how to tame them. *arXiv:2411.16171*, 2024.
- [33] Jaehui Hwang, Junghyuk Lee, and Jong-Seok Lee. Anomaly score: Evaluating generative models and individual generated images based on complexity and vulnerability. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8754–8763, 2024.

- [34] Lucas Theis, Aäron van den Oord, and Matthias Bethge. A note on the evaluation of generative models. *arXiv:1511.01844*, 2015.
- [35] Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*, pages 7176–7185. PMLR, 2020.
- [36] Vitali Petsiuk, Alexander E Siemenn, Saisamrit Surbehera, Zad Chin, Keith Tyser, Gregory Hunter, Arvind Raghavan, Yann Hicke, Bryan A Plummer, Ori Kerret, et al. Human evaluation of text-to-image models on a multi-task benchmark. *arXiv:2211.12112*, 2022.
- [37] Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. Imagereward: Learning and evaluating human preferences for text-to-image generation. *Advances in Neural Information Processing Systems*, 36:15903–15935, 2023.
- [38] Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. 2023.
- [39] David Bau, Jun-Yan Zhu, Jonas Wulff, William Peebles, Hendrik Strobelt, Bolei Zhou, and Antonio Torralba. Seeing what a GAN cannot generate. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4502–4511, 2019.
- [40] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps. *Proceedings of the International Conference on Learning Representations (ICLR)*, 2013.
- [41] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. *Proceedings of the International Conference on Machine Learning (ICML)*, 2017.
- [42] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [43] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. *Advances in Neural Information Processing Systems (NIPS)*, 2018.
- [44] Amirata Ghorbani, Abubakar Abid, and James Zou. Interpretation of neural networks is fragile. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2017.
- [45] Peter Hase and Mohit Bansal. Evaluating explainable ai: Which algorithmic explanations help users predict model behavior? *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- [46] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, et al. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav). *Proceedings of the International Conference on Machine Learning (ICML)*, 2018.
- [47] David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. Network dissection: Quantifying interpretability of deep visual representations. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [48] Thomas Fel, Agustin Picard, Louis Bethune, Thibaut Boissin, David Vigouroux, Julien Colin, Rémi Cadène, and Thomas Serre. Craft: Concept recursive activation factorization for explainability. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [49] Matthew Kowal, Achal Dave, Rares Ambrus, Adrien Gaidon, Konstantinos G Derpanis, and Pavel Tokmakov. Understanding video transformers via universal concept discovery. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

- [50] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *ArXiv e-print*, 2023.
- [51] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- [52] Thomas Fel, Victor Boutin, Mazda Moayeri, Remi Cadene, Louis Bethune, Mathieu Chalvidal, and Thomas Serre. A holistic approach to unifying automatic concept extraction and concept importance estimation. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [53] David Chanin, James Wilken-Smith, Tomas Dulka, Hardik Bhatnagar, and Joseph Bloom. A is for absorption: Studying feature splitting and absorption in sparse autoencoders. *ArXiv e-print*, 2024.
- [54] Martin Wattenberg and Fernanda B Viegas. Relational composition in neural networks: A survey and call to action. *ArXiv e-print*, 2024.
- [55] Usha Bhalla, Suraj Srinivas, Asma Ghandeharioun, and Himabindu Lakkaraju. Towards unifying interpretability and control: Evaluation via intervention. *ArXiv e-print*, 2024.
- [56] Thomas Fel, Ekdeep Singh Lubana, Jacob S Prince, Matthew Kowal, Victor Boutin, Isabel Papadimitriou, Binxu Wang, Martin Wattenberg, Demba Ba, and Talia Konkle. Archetypal SAE: Adaptive and stable dictionary learning for concept extraction in large vision models. 2025.
- [57] Arpita Chowdhury, Dipanjyoti Paul, Zheda Mai, Jianyang Gu, Ziheng Zhang, Kazi Sajeed Mehrab, Elizabeth G Campolongo, Daniel Rubenstein, Charles V Stewart, Anuj Karpatne, et al. Prompt-CAM: A simpler interpretable transformer for fine-grained analysis. *arXiv:2501.09333*, 2025.
- [58] David Bau, Jun-Yan Zhu, Hendrik Strobelt, Bolei Zhou, Joshua B Tenenbaum, William T Freeman, and Antonio Torralba. GAN dissection: Visualizing and understanding generative adversarial networks. 2018.
- [59] Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, and Ferhan Ture. What the DAAM: Interpreting stable diffusion using cross attention. 2022.
- [60] Ali Borji. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing*, 137:104771, 2023.
- [61] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with CLIP latents. *arXiv preprint arXiv:2204.06125*, 1(2):3, 2022.
- [62] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2020.
- [63] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. 2023.
- [64] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *International conference on machine learning*, pages 8162–8171. PMLR, 2021.
- [65] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *Advances in Neural Information Processing Systems*, 35:8633–8646, 2022.
- [66] Haoyu Lu, Guoxing Yang, Nanyi Fei, Yuqi Huo, Zhiwu Lu, Ping Luo, and Mingyu Ding. VDT: General-purpose video diffusion transformers via mask modeling. *arXiv preprint arXiv:2305.13311*, 2023.

- [67] Jiuniu Wang, Hangjie Yuan, Dayou Chen, Yingya Zhang, Xiang Wang, and Shiwei Zhang. ModelScope text-to-video technical report. *arXiv preprint arXiv:2308.06571*, 2023.
- [68] Yuwei Guo, Ceyuan Yang, Anyi Rao, Zhengyang Liang, Yaohui Wang, Yu Qiao, Maneesh Agrawala, Dahua Lin, and Bo Dai. AnimateDiff: Animate your personalized text-to-image diffusion models without specific tuning. *arXiv preprint arXiv:2307.04725*, 2023.
- [69] Shanchuan Lin and Xiao Yang. AnimateDiff-Lightning: Cross-model diffusion distillation. *arXiv preprint arXiv:2403.12706*, 2024.
- [70] Wenyi Hong, Ming Ding, Wendi Zheng, Xinghan Liu, and Jie Tang. CogVideo: Large-scale pretraining for text-to-video generation via transformers. *arXiv preprint arXiv:2205.15868*, 2022.
- [71] Haoxin Chen, Menghan Xia, Yingqing He, Yong Zhang, Xiaodong Cun, Shaoshu Yang, Jinbo Xing, Yaofang Liu, Qifeng Chen, Xintao Wang, et al. VideoCrafter1: Open diffusion models for high-quality video generation. *arXiv preprint arXiv:2310.19512*, 2023.
- [72] Jay Zhangjie Wu, Yixiao Ge, Xintao Wang, Stan Weixian Lei, Yuchao Gu, Yufei Shi, Wynne Hsu, Ying Shan, Xiaohu Qie, and Mike Zheng Shou. Tune-a-video: One-shot tuning of image diffusion models for text-to-video generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 7623–7633, 2023.
- [73] Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3D: High-resolution text-to-3d content creation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 300–309, 2023.
- [74] Heewoo Jun and Alex Nichol. Shap-E: Generating conditional 3d implicit functions. *arXiv preprint arXiv:2305.02463*, 2023.
- [75] Haochen Wang, Xiaodan Du, Jiahao Li, Raymond A Yeh, and Greg Shakhnarovich. Score jacobian chaining: Lifting pretrained 2d diffusion models for 3d generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12619–12629, 2023.
- [76] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- [77] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.
- [78] Stability AI. Stable Diffusion 2.0 Release, 2022. URL <https://stability.ai/news/stable-diffusion-v2-release>
- [79] Stability AI. Introducing Stable Diffusion 3, 2024. URL <https://stability.ai/news/stable-diffusion-3>
- [80] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. SDXL: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023.
- [81] Kihyuk Sohn, Nataniel Ruiz, Kimin Lee, Daniel Castro Chin, Irina Blok, Huiwen Chang, Jarred Barber, Lu Jiang, Glenn Entis, Yuanzhen Li, et al. StyleDrop: Text-to-image generation in any style. *arXiv preprint arXiv:2306.00983*, 2023.
- [82] Zhihong Pan, Xin Zhou, and Hao Tian. Arbitrary style guidance for enhanced diffusion-based text-to-image generation. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 4461–4471, 2023.
- [83] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3836–3847, 2023.

- [84] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22500–22510, 2023.
- [85] Darian Tomavsevic, Fadi Boutros, Chenhao Lin, Naser Damer, Vitomir Struc, and Peter Peer. Id-booth: Identity-consistent face generation with diffusion models. *arXiv preprint arXiv:2504.07392*, 2025.
- [86] Anton Razzhigaev, Arseniy Shakhmatov, Anastasia Maltseva, Vladimir Arkhipkin, Igor Pavlov, Ilya Ryabov, Angelina Kuts, Alexander Panchenko, Andrey Kuznetsov, and Denis Dimitrov. Kandinsky: An improved text-to-image synthesis with image prior and latent diffusion. *arXiv preprint arXiv:2310.03502*, 2023.
- [87] Junsong Chen, Jincheng Yu, Chongjian Ge, Lewei Yao, Enze Xie, Yue Wu, Zhongdao Wang, James Kwok, Ping Luo, Huchuan Lu, et al. Pixart- α : Fast training of diffusion transformer for photorealistic text-to-image synthesis. *arXiv preprint arXiv:2310.00426*, 2023.
- [88] Stability AI. DeepFloyd IF: A Powerful Open-Source Text-to-Image Model, 2023. URL <https://stability.ai/news/deepfloyd-if-text-to-image-model>.
- [89] Black Forest Labs. Announcing Black Forest Labs, 2024. URL <https://blackforestlabs.ai/announcing-black-forest-labs/>.
- [90] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020.
- [91] Jiahui Yu, Yuanzhong Xu, Jing Yu Koh, Thang Luong, Gunjan Baid, Zirui Wang, Vijay Vasudevan, Alexander Ku, Yinfei Yang, Burcu Karagol Ayan, et al. Scaling autoregressive models for content-rich text-to-image generation. *arXiv preprint arXiv:2206.10789*, 2(3):5, 2022.
- [92] Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. Deep learning scaling is predictable, empirically. *arXiv preprint arXiv:1712.00409*, 2017.
- [93] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. LAION-5B: An open large-scale dataset for training next generation image-text models. *Advances in neural information processing systems*, 35:25278–25294, 2022.
- [94] Common Crawl. Common crawl corpus. <https://commoncrawl.org> 2017.
- [95] Minwoo Byeon, Beomhee Park, Haecheon Kim, Sungjun Lee, Woonhyuk Baek, and Saehoon Kim. COYO-700M: Image-text pair dataset, 2022.
- [96] Piyush Sharma, Nan Ding, Sebastian Goodman, and Radu Soricut. Conceptual Captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In *Proceedings of ACL*, 2018.
- [97] Abeba Birhane, Sanghyun Han, Vishnu Boddeti, Sasha Luccioni, et al. Into the LAION’s den: Investigating hate in multimodal datasets. *Advances in neural information processing systems*, 36:21268–21284, 2023.
- [98] Abeba Birhane, Sepehr Dehdashtian, Vinay Prabhu, and Vishnu Boddeti. The dark side of dataset scaling: Evaluating racial classification in multimodal models. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 1229–1244, 2024.
- [99] Preethi Seshadri, Sameer Singh, and Yanai Elazar. The bias amplification paradox in text-to-image generation. *arXiv preprint arXiv:2308.00755*, 2023.

- [100] Abeba Birhane, Vinay Uday Prabhu, and Emmanuel Kahembwe. Multimodal datasets: Misogyny, pornography, and malignant stereotypes. *arXiv preprint arXiv:2110.01963*, 2021.
- [101] David Thiel. Identifying and eliminating CSAM in generative ml training data and models. *Stanford Internet Observatory, Cyber Policy Center, December*, 23:3, 2023.
- [102] Ali Shirali and Moritz Hardt. What makes ImageNet look unlike LAION. *arXiv preprint arXiv:2306.15769*, 2023.
- [103] Thao Nguyen, Samir Yitzhak Gadre, Gabriel Ilharco, Sewoong Oh, and Ludwig Schmidt. Improving multimodal datasets with image captioning. *Advances in Neural Information Processing Systems*, 36:22047–22069, 2023.
- [104] Bruno A Olshausen and David J Field. Emergence of simple-cell receptive field properties by learning a sparse code for natural images. *Nature*, 1996.
- [105] Michael Elad. Sparse and redundant representations: from theory to applications in signal and image processing. 2010.
- [106] Julien Mairal, Francis Bach, and Jean Ponce. Sparse modeling for image and vision processing. *Foundations and Trends in Computer Graphics and Vision*, 2014.
- [107] David L Donoho. Compressed sensing. *IEEE Transactions on Information Theory*, 2006.
- [108] Emmanuel J Candès, Justin Romberg, and Terence Tao. Robust uncertainty principles: Exact signal reconstruction from highly incomplete frequency information. *IEEE Transactions on Information Theory*, 2006.
- [109] Stuart Lloyd. Least squares quantization in pcm. *IEEE Transactions on Information Theory*, 1982.
- [110] Daniel D Lee and H Sebastian Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 1999.
- [111] Nicolas Gillis. Nonnegative matrix factorization. 2020.
- [112] Hui Zou, Trevor Hastie, and Robert Tibshirani. Sparse principal component analysis. *Journal of Computational and Graphical Statistics*, 2006.
- [113] Michal Aharon, Michael Elad, and Alfred Bruckstein. K-svd: An algorithm for designing overcomplete dictionaries for sparse representation. *IEEE Transactions on Signal Processing*, 2006.
- [114] Julien Mairal, Francis Bach, Jean Ponce, and Guillermo Sapiro. Online dictionary learning for sparse coding. *Proceedings of the International Conference on Machine Learning (ICML)*, 2009.
- [115] Rodolphe Jenatton, Guillaume Obozinski, and Francis Bach. Structured sparse principal component analysis. *International Conference on Artificial Intelligence and Statistics*, 2010.
- [116] Daniel A Spielman, Huan Wang, and John Wright. Exact recovery of sparsely-used dictionaries. *The Journal of Machine Learning Research (JMLR)*, 2012.
- [117] Jean Barbier and Nicolas Macris. Statistical limits of dictionary learning: random matrix theory and the spectral replica method. *Physical Review E*, 2022.
- [118] Vardan Papyan, Yaniv Romano, and Michael Elad. Convolutional dictionary learning via local processing. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [119] Alex Tamkin, Mohammad Tafseeque, and Noah D Goodman. Codebook features: Sparse and discrete interpretability for neural networks. *ArXiv e-print*, 2023.
- [120] Berk Tinaz, Zalan Fabian, and Mahdi Soltanolkotabi. Emergence and evolution of interpretable concepts in diffusion models. *arXiv:2504.15473*, 2025.

- [121] Akshay Kulkarni, Ge Yan, Chung-En Sun, Tuomas Oikarinen, and Tsui-Wei Weng. Interpretable generative models through post-hoc concept bottlenecks. *arXiv preprint arXiv:2503.19377*, 2025.
- [122] Mert Yuksekgonul, Maggie Wang, and James Zou. Post-hoc concept bottleneck models. *arXiv preprint arXiv:2205.15480*, 2022.
- [123] Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. *Advances in neural information processing systems*, 32, 2019.
- [124] Jingyi Xu, Hieu Le, and Dimitris Samaras. Assessing sample quality via the latent space of generative models. In *European Conference on Computer Vision*, pages 449–464. Springer, 2024.
- [125] Iro Laina, Yuki M Asano, and Andrea Vedaldi. Measuring the interpretability of unsupervised representations via quantized reverse probing. *arXiv:2209.03268*, 2022.
- [126] Julius Von Kugelgen, Yash Sharma, Luigi Gresele, Wieland Brendel, Bernhard Scholkopf, Michel Besserve, and Francesco Locatello. Self-supervised learning with data augmentations provably isolates content from style. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [127] Francesco Locatello, Stefan Bauer, Mario Lucic, Gunnar Raetsch, Sylvain Gelly, Bernhard Scholkopf, and Olivier Bachem. Challenging common assumptions in the unsupervised learning of disentangled representations. *Proceedings of the International Conference on Machine Learning (ICML)*, 2019.
- [128] Roland S Zimmermann, Yash Sharma, Steffen Schneider, Matthias Bethge, and Wieland Brendel. Contrastive learning inverts the data generating process. *Proceedings of the International Conference on Machine Learning (ICML)*, 2021.
- [129] Luigi Gresele, Paul K Rubenstein, Arash Mehrjou, Francesco Locatello, and Bernhard Scholkopf. The incomplete rosetta stone problem: Identifiability results for multi-view nonlinear ica. *Uncertainty in Artificial Intelligence*, 2020.
- [130] Luigi Gresele, Julius Von Kugelgen, Vincent Stimper, Bernhard Scholkopf, and Michel Besserve. Independent mechanism analysis, a new concept? *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [131] Maya Okawa, Ekdeep S Lubana, Robert Dick, and Hidenori Tanaka. Compositional abilities emerge multiplicatively: Exploring diffusion models on a synthetic task. *Advances in Neural Information Processing Systems*, 36:50173–50195, 2023.
- [132] Core Francisco Park, Maya Okawa, Andrew Lee, Ekdeep S Lubana, and Hidenori Tanaka. Emergence of hidden capabilities: Exploring learning dynamics in concept space. *Advances in Neural Information Processing Systems*, 37:84698–84729, 2024.
- [133] Ilyes Khemakhem, Diederik Kingma, Ricardo Monti, and Aapo Hyvarinen. Variational autoencoders and nonlinear ica: A unifying framework. *Proceedings of the International Conference on Machine Learning (ICML)*, 2020.
- [134] Aapo Hyvarinen, Hiroaki Sasaki, and Richard Turner. Nonlinear ICA using auxiliary variables and generalized contrastive learning. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pages 859–868. PMLR, 2019.
- [135] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *ArXiv e-print*, 2023.
- [136] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. *Transformer Circuits Thread*, 2022.

- 860 [137] Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya
861 Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. *Proceedings*
862 *of the International Conference on Learning Representations (ICLR)*, 2025.
- 863 [138] Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen,
864 Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L
865 Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers,
866 Edward Rees, Joshua Batson, Adam Jermy, Shan Carter, Chris Olah, and Tom Henighan.
867 Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer*
868 *Circuits Thread*, 2024.
- 869 [139] Senthooan Rajamanoharan, Tom Lieberum, Nicolas Sonnerat, Arthur Conmy, Vikrant Varma,
870 Janos Kramar, and Neel Nanda. Jumping ahead: Improving reconstruction fidelity with
871 jumprelu sparse autoencoders. *ArXiv e-print*, 2024.
- 872 [140] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward
873 Grefenstette, and Roberta Raileanu. Understanding the effects of rlhf on llm generalisation
874 and diversity. *arXiv preprint arXiv:2310.06452*, 2023.
- 875 [141] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and
876 Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model.
877 *Advances in Neural Information Processing Systems*, 36:53728–53741, 2023.
- 878 [142] Colin McDiarmid et al. On the method of bounded differences. *Surveys in combinatorics*, 141
879 (1):148–188, 1989.

880

881

Appendix

882

883

882

Table of Contents

884	A Experimental Setup	24
885	A.1 Observation Space	24
886	A.2 Synthesized Images	24
887	A.3  Distribution Level Analysis	24
888	A.4  Datapoint Level Analysis	25
889	A.5  Co-occurrence Analysis	25
890	B Custom RA-SAE Training Details	26
891	C Computational Resources	27
892	D Autointerpretability	28
893	E Additional Results: Model-Specific Blindspots	30
894	F Additional Results: Higher-order Blindspots with Compositional Discrepancy	32
895	G Additional Results: Blindspot Stress Testing	34
896	G.1 Bird Feeder Blindspot in Kandinsky	34
897	G.2 Glossy DVD Disc Blindspot in SD 1.5	35
898	H Additional Examples: Synthesized Images	36
899	I Concentration bounds for δ	37
900	J Monotonicity and Calibration-Free Interpretation of δ	38
901	K Stability of FID Under SAE Embeddings	39

902

903

904

A Experimental Setup

This section specifies, in full, the experimental setup underlying our analysis of four widely used generative image models—SD 1.5/2.1, Kandinsky, and PixArt—all trained on LAION-5B or on its subsets and derivatives. The complete code is available at a software repository.

A.1 Observation Space

We construct the observation space by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Owing to concerns over CSAM and other unsafe content, the original release is no longer available; we therefore adopt a substitute release of a subset of the dataset, subjected to additional filtering of unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure comprises three sequential steps: (1) loading the full LAION dataset through the Hugging Face `datasets` library; (2) validating each candidate by confirming a well-formed URL and resource availability via an HTTP HEAD request; and (3) sampling with replacement until 10,000 valid samples are obtained. This procedure yields D_G as a set of (x, t) tuples and guarantees an observation space of accessible image-text pairs—a non-trivial requirement, given that a large fraction of the original image URLs has become unavailable since release. Additional examples of synthesized images are provided in Appendix H.

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in exact one-to-one correspondence with D_G , producing triplets (x, x', t) . Concretely, for each of the 10,000 image-text pairs (x, t) in D_G , we condition the generative model g_θ on the caption t so as to synthesize the counterpart image x' .

Synthesis follows the standard text-to-image generation pipeline of each architecture, implemented with the Hugging Face `diffusers` library and executed at mixed precision (fp16). Unless stated otherwise below, all synthetic images are generated at 512×512 resolution under default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5, employed in the analysis of Sec. 5.5, coincides with the baseline SD 1.5 implementation in every respect save one: its UNet is replaced by the DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 For each evaluated model, we compute the energy difference $\delta(\cdot)$ across all 32,000 concepts, applying the sigmoid transformation with temperature $T = 0.8$ during normalization. The resulting values are rendered as log-scale density histograms with 100 bins spanning $[0, 1]$.

950 **Section 5.2** We embed the complete set of 32,000 concepts into two dimensions via UMAP applied
 951 to the sparse concept codes, coloring each point by its energy difference $\delta(\cdot)$ so as to foreground
 952 both suppressed and exaggerated blindspots. To quantify cross-model consistency, we then compute
 953 pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors of every model pair, presenting the
 954 results as both scatter plots and correlation matrices. This analysis discloses whether blindspots cluster
 955 in conceptual space and cleanly disambiguates model-specific from universal blindspot patterns.

956 **Section 5.3** We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually examine the
 957 extrema at both ends of the ranking. For suppressed blindspots we select concepts with $\delta(\cdot) < 0.1$;
 958 for exaggerated blindspots, those with $\delta(\cdot) < 0.9$. The presented examples are annotated with textual
 959 concept descriptions obtained by inspecting each concept’s most activating images together with its
 960 spatial attention patterns. Our ongoing efforts to automate this concept interpretation are detailed in
 961 Appendix D

962 **Section 5.5** We compare SD 1.5 with and without DPO as follows: for each image pair (x, x') ,
 963 we compute the L2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$,
 964 applying a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences prior
 965 to averaging. This yields datapoint-wise energy differences that quantify precisely how far each
 966 generated image departs from its natural counterpart in concept space. The differences are then
 967 visualized as overlapping histograms that directly contrast the two model variants.

968 **Section 5.6** For each concept c_k , we count its empirical frequency $\|Z_{:,i}\|_0$ —the number of non-
 969 zero activations across the natural dataset—and apply a sigmoid normalization with temperature
 970 $T = 0.4$ to the energy differences $\delta(\cdot)$. The relationship is displayed as a scatter plot with empirical
 971 concept frequency on the x-axis and the sigmoid-transformed energy difference on the y-axis; point
 972 size is proportional to activation frequency, and point color encodes the magnitude of the energy
 973 difference.

974 A.4 • Datapoint Level Analysis

975 **Section 5.4** For each image pair (x, x') , we compute the L2 norm of the difference between
 976 their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, yielding a single scalar measure of conceptual
 977 divergence per pair. Ranking the samples by this divergence exposes the two failure regimes at its
 978 extremes: minimal divergence flags potential memorization artifacts, whereas maximal divergence
 979 flags substantial conceptual failures. The ranking thereby affords targeted qualitative inspection of
 980 each specific failure mode.

981 A.5 • Co-occurrence Analysis

982 **Appendix F** For both the natural and the synthesized data, D_G and D_{g_0} , we analyze concept
 983 co-occurrence through the co-activation matrix $Z^T Z$, which encodes the pairwise correlations in
 984 concept usage. We then extract the dominant conceptual directions by eigendecomposition and
 985 assess the alignment between the natural and synthetic co-occurrence structures via cosine-similarity
 986 heatmaps between the top-100 eigenvectors of the two matrices. Each such 100×100 similarity
 987 matrix is displayed as a square heatmap: a perfect diagonal structure would signify identical principal
 988 concept axes, whereas off-diagonal mass exposes rotations and mismatches in the compositional
 989 geometry.

B Custom RA-SAE Training Details

This section documents, in full, the training details of our custom archetypal SAE (RA-SAE) [56], open-sourced at a software repository.

Dataset. The autoencoder is trained on the complete ImageNet-1k training split (≈ 1.28 M RGB images). Each image is encoded into 261 visual tokens by DINOv2 [135], and these tokens are fed to the SAE without class or position embeddings. The training corpus therefore comprises $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$ tokens.

Dictionary. The dictionary spans 32,000 concept dimensions. Sparsity is enforced by top- k masking with $k = 5$: for each input, all activations outside the five largest are set to 0. Weights are initialized via Xavier/Glorot, and training proceeds at mixed precision (fp16), with the final ten epochs conducted at full precision to sharpen convergence.

Optimizer and schedule. The model is trained for 50 epochs with the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and weight decay 10^{-5} . The learning rate follows a linear warm-up over the first 5% of steps and a subsequent cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective is an MSE reconstruction loss augmented by an auxiliary term, weighted by $\lambda = 10^{-5}$, that penalizes activations which never enter the top- k set.

1006 C Computational Resources

1007 This section itemizes the GPU resources consumed by the training and experiments reported in this
1008 paper, amounting in total to approximately 202 GPU-hours on NVIDIA H100 and H200 hardware.

1009 **RA-SAE.** Training required approximately 24 GPU-hours on three NVIDIA H100s.

1010 **Synthesized images.** Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator
1011 when distributed across four NVIDIA H200 GPUs; aggregated across five generators, this amounted
1012 to approximately 100 GPU-hours on the H200.

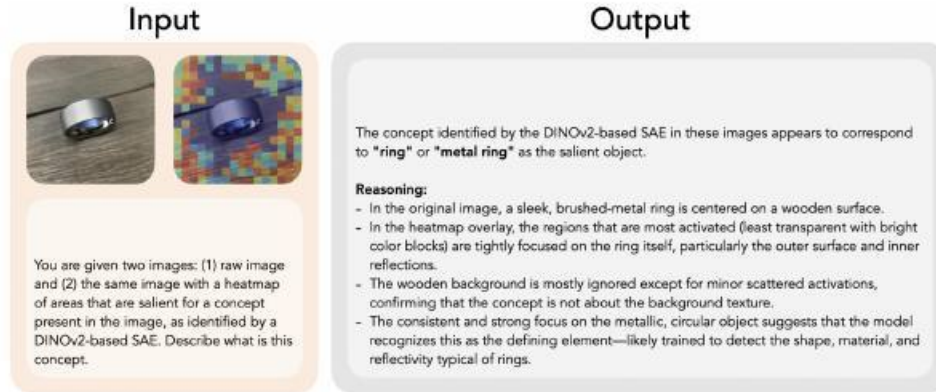
1013 **• Distribution-level analysis.** Extracting distribution-level energy differences took about 3 hours
1014 per generator on a single NVIDIA H200 GPU (~ 15 GPU-hours in total).

1015 **• Datapoint-level analysis.** Computing datapoint-level energy differences, ranking concepts per
1016 datapoint, and ranking datapoints per concept together took approximately 3 hours per generator on
1017 one H200 GPU (~ 15 GPU-hours in total).

1018 D Autointerpretability

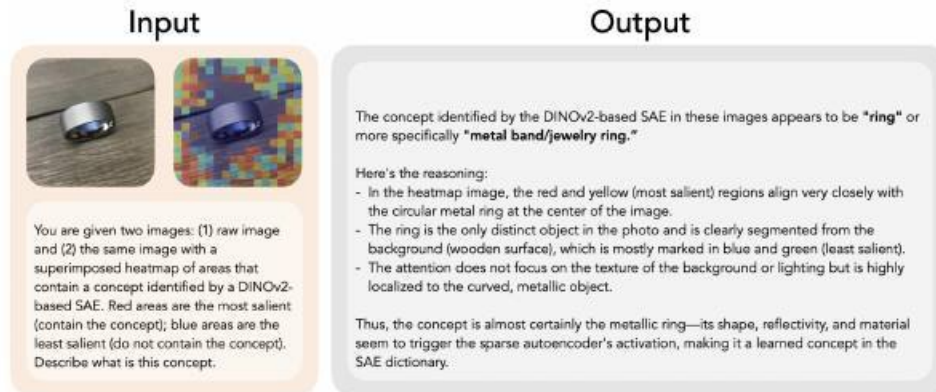
1019 Toward automating the interpretation of the concepts identified by the RA-SAE, we conducted
 1020 a qualitative evaluation of the capacity of Vision-Language Models (VLMs) to describe them.
 1021 Specifically, we queried ChatGPT 4o through the OpenAI API using a range of prompts and image
 1022 formats, each derived from an image x'' in which the target concept was salient, and recorded the
 1023 VLM’s description. This section reports the findings of that exploration. For evaluation we permitted
 1024 the VLM to emit its full reasoning; a scalable deployment would suppress this reasoning through a
 1025 minimal prompt modification requesting the description alone.

1026 We first supplied the VLM with two images: a raw version of x'' and a version with a superimposed
 1027 heatmap localizing the intensity of the concept:



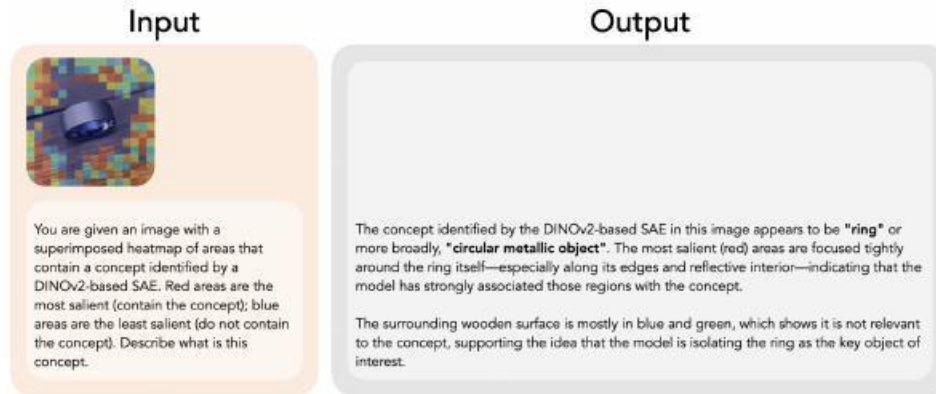
1028 As the example shows, the VLM failed to interpret the heatmap correctly: rather than describing the
 1029 wood texture in the background—the genuinely salient region—it described the foreground ring, to
 1030 which the RA-SAE assigned no activation whatsoever.

1031 To ascertain whether this failure was merely an artifact of prompting, we repeated the experiment
 1032 with an expanded prompt that explained the heatmap in fuller detail:



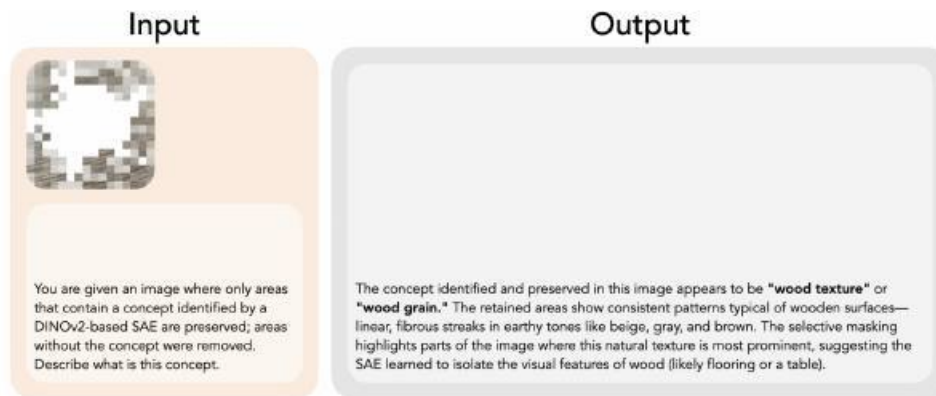
1033 This modification did not resolve the issue: the VLM once again identified the ring as the prominent
 1034 concept. Several further prompting strategies—including ones that described the heatmap in still
 1035 greater depth, for instance by supplying the code used to generate it—produced comparable results.

1036 We next provided the VLM with the heatmap-superimposed x'' alone, accompanied by a similar
 1037 prompt explaining how the heatmap is to be interpreted:



1038 Once again the VLM failed to identify the wooden background and instead fixated on the ring;
 1039 varying the prompting strategy for this image format left the results essentially unchanged.

1040 These observations lead us to conclude that the heatmap alone is insufficient to direct the VLM's
 1041 attention to the salient region, since the presence of less relevant regions distracts the model. To
 1042 eliminate this distraction, we replaced the heatmap-superimposed x'' with an alpha-masked version
 1043 of x'' :



1044 As the example shows, the VLM now correctly identified wood texture as the dominant concept,
 1045 and we observed consistent success across a broad range of concepts and images.

1046 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
 1047 evaluation. We stress, however, that this evaluation is bounded by its qualitative character—owing to
 1048 the absence of ground-truth annotations—and by its focus on a single VLM. We anticipate that future
 1049 work on the autointerpretability of SAE concepts will build directly upon and extend this analysis.

1050 E Additional Results: Model-Specific Blindspots



Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize precisely where the concept is expressed.

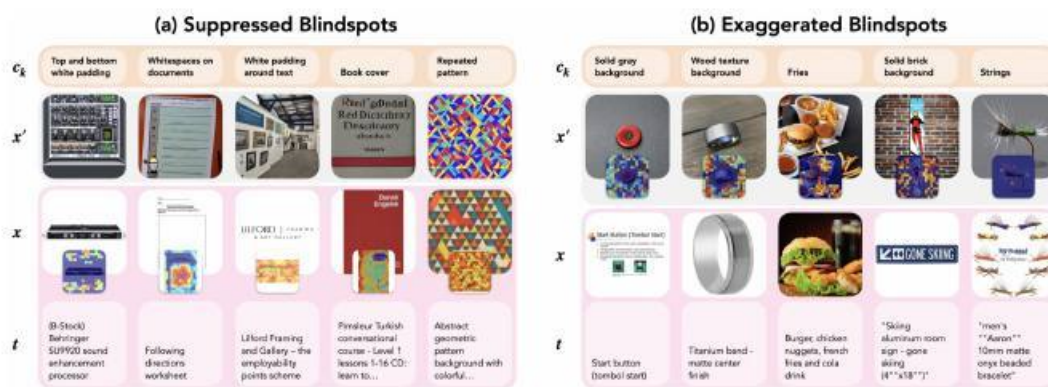


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize precisely where the concept is expressed.

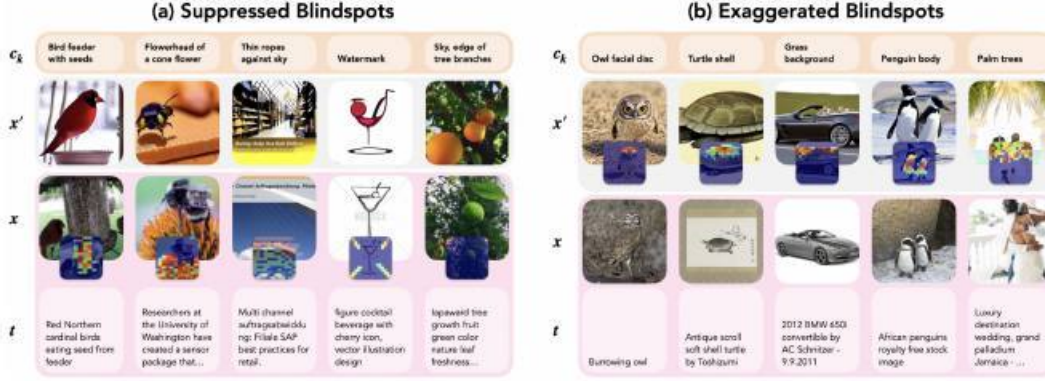


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize precisely where the concept is expressed.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize precisely where the concept is expressed.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Our analysis has thus far centered on individual concept activations. Visual scenes, however, are rarely composed of isolated concepts; they are organized instead by rich co-occurrence patterns that encode compositional semantics. We now examine whether, and to what extent, generative models capture this higher-order structure, analyzing the co-activation matrix $Z^T Z$, whose entries are the pairwise correlations in concept usage.

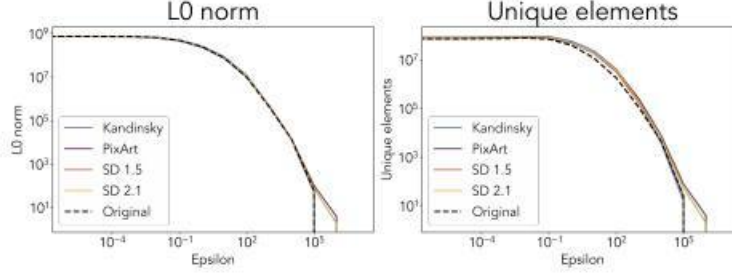


Figure 18: • **Sparsity and structural divergence.** Left: the ℓ_0 norm of the co-occurrence matrix ZZ^T as a function of the threshold ϵ , indicating how many entries remain active in each model. Right: the number of unique entries in the synthesized distribution relative to the natural distribution. Every evaluated model faithfully preserves the global sparsity structure while diverging in its specific activation content.

Notably, at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18(left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded across a range of ϵ values—tracks closely between the natural and synthesized distributions for every model. The gross connectivity of the conceptual graph—that is, which concepts co-activate at all—is thus well preserved. Formally, $Z^T Z$ admits interpretation as the adjacency matrix of a weighted, undirected graph over concepts whose edge weights encode co-activation strength across the dataset.

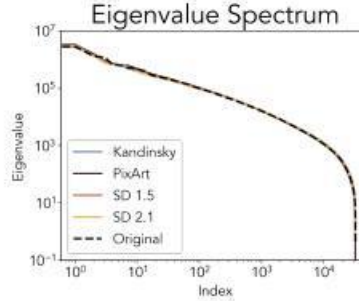


Figure 19: • **Spectral structure of co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices ZZ^T for each model and for the natural distribution. Every evaluated model faithfully reproduces the heavy-tailed decay characteristic of the natural distribution.

The specific content of these co-activations, however, diverges: as Fig. 18(right) shows, a substantial fraction of the entries in the model-generated $Z^T Z$ are absent from the natural baseline. The capacity for compositionality is thus retained even as the identity of the active pairings shifts—a shift that plausibly reflects model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 presents the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. Every spectrum exhibits heavy-tailed, power-law-consistent decay, which establishes that generative models preserve both the overall rank structure and the allocation of variance across conceptual dimensions.

We further examine the alignment of the dominant conceptual directions via cosine-similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). While every model exhibits partial diagonal alignment—confirming a substantial

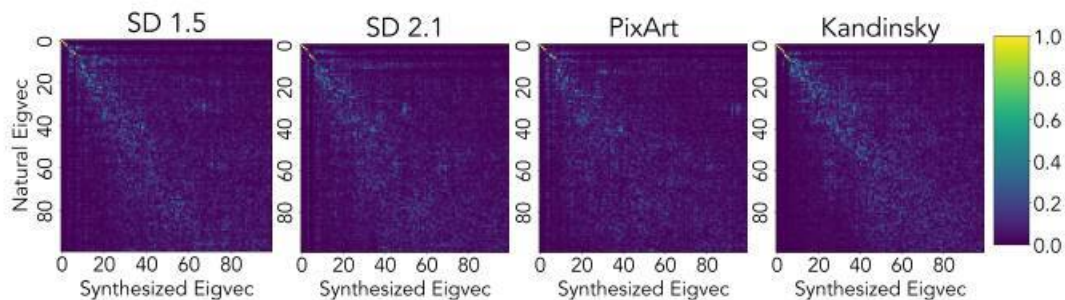


Figure 20: ● **Concept basis similarity.** Cosine-similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure signals alignment of the dominant conceptual directions, whose degree varies systematically across the four models.

1076 overlap in the principal concept axes—the off-diagonal entries expose rotations and mismatches in
 1077 the higher modes, marking clear and structured deviations in compositional geometry.

1078 Together, these findings establish a nuanced yet coherent picture: diffusion models approximate the
 1079 global shape of concept co-activation remarkably well, yet deviate in subtle and structured ways once
 1080 examined through the spectral lens. Such higher-order discrepancies furnish a plausible mechanism
 1081 for the observed failures in generating coherent multi-object scenes and relational concepts.

1082 G Additional Results: Blindspot Stress Testing

1083 To stress-test the blindspots identified by our method, we assembled a range of prompts describing
 1084 each blindspot, used them to generate many images, and contrasted the outputs of the models in
 1085 which the concept was flagged as a blindspot against those of the models in which it was not.

1086 Concretely, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1087 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1088 *text-to-image model. Return these as a list of strings in Python.* We generated five images per
 1089 prompt and analyzed them with our custom RA-SAE model (Appendix B), which ranked them by the
 1090 intensity with which the target concept appeared; all images were subsequently reviewed manually to
 1091 determine whether the blindspot was successfully depicted.

1092 As the following examples of suppressed concepts illustrate, although fragments of the target concept
 1093 occasionally surfaced—for instance a holder or string for the bird feeder blindspot, or a round
 1094 hole for the glossy DVD disc blindspot—the models systematically failed to render the complete
 1095 concept. This outcome directly corroborates our method’s assessment and thereby validates the stress
 1096 test as an independent check on the diagnostic.

1097 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from a variety of prompts involving the bird feeder concept during stress testing. Our method flags this concept as a suppressed conceptual blindspot in Kandinsky, and the generations bear this out precisely: the model proves unable to render it. SD 1.5, for which the concept was not flagged as a blindspot, generates it successfully—exactly as the diagnostic predicts.



Figure 22: Images generated from a variety of prompts involving the `glossy DVD disc` concept during stress testing. Our method flags this concept as a suppressed conceptual blindspot in SD 1.5, and the generations bear this out precisely: the model proves unable to render it. Kandinsky, for which the concept was not flagged as a blindspot, generates it successfully—again matching the diagnostic’s prediction.

1099 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pairs from LAION-5B, each displayed alongside the images synthesized from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1100 I Concentration bounds for δ

1101 In our experiments we estimate $\delta(k)$ from $n = 10,000$ paired samples per concept. This sample
 1102 budget is deliberately modest, which raises the natural question of whether it suffices for reliable
 1103 estimation. We resolve the question decisively in the affirmative, deriving a concentration bound on
 1104 the empirical estimator $\hat{\delta}_n(k)$ via McDiarmid’s inequality [142]. The bound is tight and establishes
 1105 that even a comparatively small sample yields fast, accurate estimates of concept bias.

1106 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1107 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1108 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1109 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1110 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1111 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1112 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1113 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1114 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1115 which proves the claim. □

1116 In practice, the concept scores $\delta(k)$ are sparse: the majority concentrate near zero and only a few
 1117 attain values as large as 10. The concentration bound guarantees that even at the largest observed
 1118 biases, a sample of $n = 10,000$ yields estimates $\hat{\delta}_n(k)$ that deviate from the true value by no more
 1119 than a small ε with high probability. This rigorously justifies our sampling strategy and confirms that
 1120 accurate bias measurements are fully attainable from limited data.

1121 J Monotonicity and Calibration-Free Interpretation of δ

1122 Our objective when analyzing blindspots is to rank concepts by the severity of their generative bias,
 1123 a purpose the score $\delta(k)$ serves directly in practice. One might reasonably ask whether this score
 1124 distorts the ranking relative to more elementary quantities such as the energy gap or the odds ratio.
 1125 The following result settles the question conclusively: $\delta(k)$ is a strictly increasing reparameterization
 1126 of both, and hence inherits their ordering exactly, so that ranking concepts by $\delta(k)$ demands no
 1127 calibration whatsoever.

1128 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1129 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1130 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1131 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1132 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1133 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1134 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1135 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1136 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1137 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1138 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1139 concepts.

1140

□

1141 Ranking concepts by $\delta(\cdot)$ is therefore strictly equivalent to ranking them by the energy gap or by the
 1142 conceptual generation odds ρ_k : no calibration is required, and all three quantities induce one and the
 1143 same total ordering over concepts.

K Stability of FID Under SAE Embeddings

In this section we establish an exact quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID computed after applying a possibly overcomplete SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, we assume D to have orthonormal columns but not necessarily to be square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We begin by recalling an elementary fact: when D is orthogonal and square ($k = d$), it acts as an isometry between A and Z , so that FID is preserved exactly. This regime is unrealistic, however, and we therefore turn to the overcomplete case, in which we bound the FID distortion by the extreme singular values of D . We work throughout with the Wasserstein-2 metric $\mathcal{W}_2$, exploiting the fact that FID is precisely $\mathcal{W}_2^2$ specialized to Gaussians.

For a probability measure μ on $\mathbb{R}^d$, we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Let $\sigma_{\min}$ and $\sigma_{\max}$ denote the minimal and maximal singular values of D —equivalently, the square roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically one typically finds $\sigma_{\min}, \sigma_{\max} \approx 1$, though the proof relies on no such assumption. We first record a simple lemma for the non-overcomplete case.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This case is nonetheless unrealistic, since SAEs rely on overcompleteness precisely in order to extract meaningful, interpretable concepts. In the overcomplete regime D is no longer orthonormal, yet it can remain column-orthonormal; it is exactly this property that we leverage to bound the distortion in terms of the extremal singular values of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1178 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1179 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1180 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1181 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1182 finishes the proof. $\square$

1183 In essence, Theorem 6 establishes that applying a column-orthogonal overcomplete SAE dictionary
 1184 distorts the Fréchet Inception Distance by no more than the square of its extremal singular values.
 1185 When D is nearly orthogonal—and empirically we consistently observed $\sigma_{\min}, \sigma_{\max} \approx 1$ —the result
 1186 guarantees that FID measured in the SAE feature space is, for all practical purposes, indistinguishable
 1187 from the canonical FID. $\square$

1188